Sustainability of inland water transport on the Rhine-Main-Danube Waterway

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Abstract*—Sustainability significantly enhances the benefits of inland waterways in various ways. Environmentally, inland waterways produce fewer emissions than road and rail transport, which helps reduce air and noise pollution and mitigates climate change by lowering CO_2 emissions therefore plays a cruicil role in cognitive mobility. Economically, transporting goods via inland waterways can be more cost-effective due to lower fuel consumption and maintenance costs, leading to reduced transportation expenses and increased economic competitiveness when ration decision allows such transport mode. Additionally, waterways play a crucial role in managing water levels, thereby reducing the risk of floods and droughts by storing excess water during heavy rains and releasing it during dry periods, so inland navigation is a tool for cognitive water management as well. Inland water transport, a cornerstone of the European transport system, is especially vital in transporting bulk, general, and liquid cargo by vessels for longer distances. Sustainability and inland water transport are interconnected, and their interconnections can be seen in various fields, such as reducing greenhouse gas emissions by implementing alternative fuels, efficient use of resources during transport and transhipment cargo operations, social and economic sustainability seen especially in the passenger transport, or innovation and technology through materials used in shipbuilding. The Rhine-Main-Danube waterway, the busiest European river for freight transport, is a critical player in this system. For inland water transport to be competitive compared to other modes of transport, various changes regarding the construction of ships or the use of new fuels must be made to make this transport sustainable in the transport system. This article explores the sustainability of inland water transport in the Rhine and Danube regions and the implementation of materials used in shipbuilding that increase vessel speed and transport capacity.

Keywords—sustainability, inland water transport, Rhine-Main-Danube waterway, freight transport

I. INTRODUCTION

Transport sustainability means a transport system designed to be ecologically, economically, and socially acceptable in the short and long term. Sustainable transport considers the needs of present generations while not compromising the ability of future generations to meet their own needs. The sustainability of inland water transport can be seen in various ways, such as reducing greenhouse gas emissions by implementing alternative fuels (hydrogen, electric energy, biofuels), optimization of handling operations in the ports during transport, transhipment, and cargo storage, social and economic sustainability seen especially in the passenger transport, or innovation and technology through different materials used in shipbuilding. [1]

Inland water transport, or inland navigation, is more than just a mode of transport. It is a green and economical solution that stands out in the world's transportation system. Inland vessels (motor cargo vessels or pushed convoys) are crucial for transporting bulk, general, or liquid cargo for longer distances. Historical, geographical, or economic factors determine its position in the transport system.

Europe has about 29 thousand kilometres of inland waterways on which inland water transport is carried out by inland vessels (motor cargo vessels or pushed convoys). The Rhine – Main – Danube waterway is the busiest European waterway from the point of view of freight transport. Not only does it connect the North and Black Seas, but it also connects fifteen European countries. Although the Danube is longer than the Rhine and flows through more European countries, the Rhine is a busier European river from the point of view of freight and passenger transport [2], [3].

One way to speed up this mode of transport and make it more sustainable in the transport system is to implement some changes in the materials used to ship new inland vessels.

The research question of this paper is: What factors influence the sustainability of inland navigation?

The paper structure is as follows. The second chapter describes the geographical area and the infrastructure. The third chapter describes the logistic performance of European inland waterways. The fourth chapter investigates the materials used in shipbuilding. The fifth chapter concludes the paper.

II. European waterway network and the Rhein-Main-Danube

The country must have a very dense river network for proper water transport. Waterways play an essential role in transporting people or cargo and the economy of individual countries. They connect seaports on the coasts of oceans or seas with industrial centres inland.

The European continent has 29 thousand kilometres of waterways, of which 14 thousand km are waterways of international importance (class IV to VII according to AGN).

The European network of waterways can be divided into two categories:

- a unified European shipping network, which includes the countries of Western, Central, and Eastern Europe, including Ukraine and the European part of the Russian Federation,
- isolated parts of the European shipping network, including Great Britain and Ireland, Scandinavia (Norway, Sweden, Finland), northern Italy, and the Iberian Peninsula.

One of the navigable waterways within the European waterway network is the Rhine – Main – Danube waterway. This waterway connects the North Sea with the Black Sea, is more than 3,500 kilometres long, and flows through fifteen European countries [2], [3]

A. *The Rhine River*

The Rhine originates in the Swiss Alps, and in its upper course, it is the border river of Switzerland with Austria, after flowing through Lake Constance and with Germany. From the Swiss city of Basel (Basel) – the highest Rhine port – it is a canalized and energy-utilized French-German border stream. After further flowing through Germany and the Netherlands, it empties into the North Sea with the French-Belgian-Dutch river Maas. Together with the navigable right-hand tributaries Neckar, Main (Main), and Lahn and the left-hand tributary Mosel (Moselle), as well as the connecting branches to the canals in the North of Germany, it is the basic waterway of Western Europe.

The European Agreement on Main Inland Waterways of International Importance (AGN) classifies the Rhine waterway as part of the most important Western European diametral E10 – the route connecting the North and Mediterranean Seas via the Rhine-Rhône Canal and the French Saône and Rhôn rivers.

Navigation on the Rhine is regulated by the Mannheim Act of 1855. Foreign vessels may sail on the Rhine only under conditions set by the Central Rhine Navigation Commission (the so-called Rhine Commission). It mainly concerns technical requirements for vessels and tests for pilots. Unlike the Danube practice, all decisions of the Rhine Commission are legally binding for the member states [2], [3].

B. *The Main River*

Near the city of Mainz, the most crucial right-hand tributary, the Main, flows into the Rhine. This river, with 34 energy-navigable stages, originates in the mountains near the German-Czech border and is navigable for approximately a length. It is 340 km from Bamberg, where the Main-Danube canal connects the Main via Nuremberg, and the continent's watershed with the Danube begins in the course of its left-hand tributary Regnitz [2], [3].

The Main-Danube Canal

The construction of the Main-Danube Canal started in 1960 when its northern part from Bamberg began. It was completed in 1992 when it was also opened for commercial shipping. The canal connected the North and Black Seas, and a total drop of 815 m is overcome when sailing from the North to the Black Sea. The total length is 171 km, and it has 16 locks. Here, we find the highest lock in Europe, located on the Hilpoltstein step at 406 meters above sea level. The main port on the canal is the port of Nuremberg [2], [3].

C. *The Danube River*

After the opening of the Main - Danube Canal in 1992, the Danube River was connected to the waterways of Western Europe via the Main River - a right-hand tributary of the Rhine River. The Danube originates as the confluence of the two German mountain rivers, Breg and Brigach, in the Black Forest near Donaueschingen. The total length of the Danube, from the confluence to the mouth of the Black Sea, is 2,845 kilometres (some sources say 2,580 kilometres). The length of the Danube, from its longer tributary (Breg), is up to 2,888 kilometres.

Commercial navigation on the Danube is carried out from the German city of Kelheim (km 2,414.72). The Danube flows through ten European countries (Germany, Austria, Slovakia, Hungary, Croatia, Serbia, Romania, Bulgaria, Moldova and Ukraine) and four capital cities (Vienna, Bratislava, Budapest and Belgrade). This number can be included among the waterways that flow with the most significant number of countries. Shortly before the mouth of the Black Sea, it forms the so-called Danube Delta, which consists of three branches. The middle arm (Sulin arm) is used for commercial shipping [4].

III. Freight transport on the Rhine-Main-Danube waterway

The inland water transport has faced significant economic, geopolitical, and environmental challenges [5] in the last few years. However, it is still an essential mode of transport in the European transport system, especially for transporting bulk, oversized, and overloaded cargo.

This sector has been mainly affected by the slowdown of the economic chain, high inflation, supplier disruptions, and energy crises related to the pandemic and the conflict between Russia and Ukraine. Low water levels, particularly on the Rhine between July and August 2022, badly influenced navigation and transport capacity.

The fleet of European inland vessels includes nearly 10,000 vessels registered in Rhine countries and 3,500 in Danube countries. Employment in the inland water transport sector has fluctuated, with the COVID-19 pandemic causing significant declines. However, the number of companies has grown since 2013 [5].

A. *Freight transport on the Rhine*

The Rhine is a critical river in Western European countries, especially Germany and the Netherlands. Freight transport is highly developed and strategically significant for the European trade market. 2022 economic and environmental challenges influenced freight transport on the Rhine, including transporting mineral oils, chemical products, sand, gravel, stones, and containers—low water levels on the Rhine limited navigation, especially in July and August 2022. Total transport volume reached 292 million tons, which was a decrease of 6.8% compared to the previous year.

The traditional Rhine, between Basel and the border between Germany and the Netherlands, has a more even distribution of commodities such as coal and steel industry final products. The Lower Rhine in the Netherlands has a higher share of chemical products and container transport than

the traditional Rhine. There is also more intensive transportation of sand, gravel, and stones [5].

B. *Freight transport on the Danube*

The Danube is the second-longest river in Europe and is essential for ten Danube countries. In 2022, various macroeconomic and geopolitical factors affected cargo movement on the Danube, including the war conflict in Ukraine and subsequent disruptions in supply chains.

In 2022, cargo transport on the Danube decreased in all segments, especially in the upper and middle parts of the river. Transport volumes were significantly reduced, with some segments down by up to 80%. The Lower Danube recorded increased transport due to importing Ukrainian cereals via an alternative route—the Danube-Black Sea Canal, which links the Danube with the Black Sea.

Agricultural products (cereals and agrobulk) recorded the largest decrease, especially in the upper and middle parts of the Danube. Grain exports from Ukraine were severely disrupted, leading to a significant drop in shipments. Chemicals and mineral oils declined due to supply chain disruptions and high prices [5].

C. *The differences between the Rhine and Danube*

The Rhine and Danube are key inland waterways in Europe, but each has its specific challenges and opportunities. Differences in infrastructure, investment, and geopolitical influences lead to differences in cargo transportation. Investment in infrastructure and adaptation to climate and geopolitical changes continue to ensure waterways' future resilience and efficiency.

High inflation, the energy crisis, and low water levels during the summer months affected freight transport on the Rhine, which limited sailing capacity. Developed infrastructure and a dense network of industrial and container terminals in the lower part of the Rhine contributed to a higher volume of transport. The Rhine transports a broader range of commodities, including chemicals, mineral oils, and containers. The Danube is more focused on transporting agricultural products, which makes it more vulnerable to geopolitical changes.

The conflict in Ukraine significantly affected transport on the Danube, disrupting the export of cereals and other agricultural products. The Lower Danube recorded increased traffic due to the export of Ukrainian cereals via alternative routes to the Black Sea. It also has less developed infrastructure and less waterway modernization investment than the Rhine [6].

IV. Materials used in shipbuilding

One way to ensure sustainable inland water transport on the Rhine-Main-Danube waterway is to use materials that increase the speed of inland vessels, transport capacity, and lifetime. Different materials (steel, aluminum, polymers) are used in shipbuilding, and each has advantages and disadvantages.

A. *Strengths and weaknesses of steel*

Steel has long been the preferred material for shipbuilding due to its strength and longevity. The main quality of steel lies in its high tensile strength, which allows it to withstand significant forces without deforming. Ships must have this property to cope with waves, strong winds, and other natural forces at sea.

Steel also has high impact resistance. A steel ship is less vulnerable to serious structural damage in a collision or impact than other, more fragile materials. Additionally, steel's ability to resist abrasion is an important advantage, especially for parts of the ship that are subject to constant friction, such as the hull.

Another significant advantage of steel is its ease of repair. Well-established welding technologies allow for relatively simple and cost-effective repairs, extending the life of steel vessels. Additionally, steel is abundant and relatively inexpensive compared to other advanced materials, making it an economically viable option for building large commercial and military ships.

However, steel also has significant disadvantages, mainly its susceptibility to corrosion. In a marine environment, constant exposure to salt water can cause rapid degradation of unprotected steel. This requires regular protective coatings and anti-corrosion treatments, increasing upkeep and maintenance costs.

Steel is also relatively heavy, limiting ships' speed and fuel efficiency. Steel ships require more power to move, which increases fuel consumption and operational costs. Additionally, the high weight reduces payload capacity, which can disadvantage certain ships, such as cargo ships.

Finally, although steel is strong, it is less flexible than some modern materials. This means that steel may be more susceptible to cracking under extreme or repeated stress, requiring regular inspections and repairs to ensure the safety and structural integrity of the vessel [7], [8], [9].

B. *Strengths and weaknesses of aluminium*

Aluminium is growing in popularity in shipbuilding, especially for smaller vessels and high-end yachts. Lightness is one of aluminium's main advantages. An aluminium boat can be significantly lighter than a steel boat of the same size, resulting in increased speed and higher fuel efficiency. The decrease in weight also leads to an increase in payload, which benefits transport ships.

Aluminum has remarkable corrosion resistance, particularly in a marine environment. Unlike steel, aluminium is coated with an oxide layer that protects it from deterioration caused by exposure to salt water. This allows for a significant reduction in maintenance and servicing expenses, making aluminium attractive for marine applications.

In terms of processing and formability, aluminium is easier to cast and weld than steel. This allows for more complex and innovative designs tailored to the specific needs of shipbuilders. Aluminum's flexibility also allows repairs to be made more efficiently, with welding methods like those used for steel, but often with lighter, less expensive equipment.

Despite its many advantages, aluminium also has notable disadvantages. Its tensile strength is lower than steel, meaning it may be less sturdy under extreme forces. This limits its use for huge ships or those carrying weighty loads.

Aluminium is also more expensive than steel. Initial manufacturing costs may be higher due to the price of raw aluminium and the specialized manufacturing processes required to work with this material. Although maintenance costs are reduced due to corrosion resistance, the high initial cost can be a barrier for some shipbuilders.

Finally, although aluminium is corrosion resistant, it is susceptible to galvanic corrosion when in contact with other metals, such as steel. This requires additional care in design and construction to avoid electrolytic contact points, which can add complexity and cost to the manufacturing process [7], [8], which could be the disadvantages of such cognitive and rational material selection [9].

C. *Strengths and weaknesses of polymers*

Due to their unique characteristics, polymers, particularly Fiber-Reinforced Polymers (FRP), are increasingly used in shipbuilding. Glass fibre composites (GFRP—Glass Fiber Reinforced Polymers) and carbon fibre composites (CFRP—Carbon Fiber Reinforced Polymers) are commonly used polymers.

One of the main advantages of polymers is their lightness. Boats made from polymers can be considerably lighter than steel or aluminium ones. This weight reduction improves speed and fuel efficiency, allowing ships to burn less fuel and carry more payload. For example, a GFRP recreational boat can sail faster and consume less fuel than a steel boat of equivalent size.

Polymers also provide excellent corrosion resistance. Unlike metals, polymers do not rust and are naturally resistant to salt water. This significantly reduces service and maintenance costs, as polymer vessels do not need the frequent anti-corrosion treatments required for steel vessels.

Polymers are very advantageous in terms of design flexibility. Moulding techniques allow the creation of complex and optimized shapes adapted to shipbuilders' needs. This flexibility allows innovative designs to improve the vessel's hydrodynamic performance and efficiency.

Fiber-reinforced polymers, particularly CFRP, provide high tensile strength and exceptional rigidity. This makes building solid and durable structures possible while maintaining minimal weight. CFRP vessels can, therefore, support significant loads and withstand harsh maritime conditions.

Despite their many advantages, polymers also have notable disadvantages. The main disadvantage is their high cost. Composite materials, especially carbon fibre-based ones, can be much more expensive than traditional metals. The manufacturing and processing processes of reinforced polymers also require specialized equipment and skills, which further increases production costs.

Polymers may also be less impact-resistant than metals. Although fibre-reinforced composites are solid, they may be more susceptible to cracking or delamination under significant impact forces. This may limit their use in applications where collisions and impacts are expected.

Another disadvantage is the difficulty of repair. Damaged polymer structures can be more challenging to repair than steel or aluminium ones. Repairs often require specialist skills and materials, which can be costly and time-consuming. Additionally, detecting internal damage in composites can be more complex, requiring advanced inspection techniques.

Polymers are also sensitive to high temperatures. Their mechanical strength can decrease significantly at elevated temperatures, posing problems in marine environments exposed to extreme temperature variations. To maintain their performance under these conditions, specially formulated polymers or protective treatments must be used [7], [8].

V. Conclusion

Inland water transport is a key player in the European Union's transport system; economic, natural, and historical factors determine its position. Its sustainability is based on some pillars and includes aspects such as innovation technologies, changes in materials used in shipbuilding, infrastructure improvement of waterways and inland canals, and legislation measures for supporting ecological solutions.

The article is focused on the sustainability of this mode of transport on the Rhine-Main-Danube waterway, which is a crucial waterway from the point of view of freight transport for the countries of the European Union and differences between the Rhine and Danube regions from the point of view of freight and passenger transport. The differences have been determined by political and economic factors such as the war conflict in Ukraine, changes in transport flows for some types of cargo (agriculture products), and the blockade of Ukrainian ports. From the side of the Russian Federation, there was an increase in prices for some commodities, such as coal or iron ore, due to this war conflict.

Implementing new materials in constructing inland vessels is an important step towards innovation and upgrading inland water transport. Materials used for decades are being replaced by easier, more flexible, and resilient materials that make vessels faster. One rational step towards to sustainable goals with the tools of cognitive mobility.

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